

Python Fundamentals

Lab 3

This lab covers Pandas, data cleaning, and data analysis. (supermarket.csv)

● Data Profiling

Before cleaning the data, profile the dataset and provide your notes.

Find the number of rows and columns.

- 1) Display the column names and data types.
- 2) Check the number of null values in each column.
- 3) Check the number of duplicate rows.
- 4) Display basic statistical information.
- 5) Identify any unnecessary or suspicious data.
- 6) Write your observations before handling the data.

● Data Cleaning

- 1) Rename the columns using clear and readable names.
- 2) Handle missing values when necessary.
- 3) Remove duplicate rows.
- 4) Remove unnecessary columns.
- 5) Make sure all columns have the correct data types.
- 6) Sort the data where appropriate.
- 7) Create a new 3 columns Day, Month, Year
- 8) Create a new column called satisfied based on rating.

Satisfied rule: $\text{rating} \geq 7 \rightarrow \text{Satisfied}$ Else Not Satisfied

● Data Analysis

The business has some questions that we need to answer. You can use Matplotlib, Seaborn, or Streamlit to make the results more visual and easier to understand.

- 1) Provide the total revenue generated by the supermarket.
- 2) Provide the total revenue by city and identify the city with the highest revenue.
- 3) Provide the total profit by branch and identify the most profitable branch.
- 4) Provide the total revenue and profit by product category and identify the top-performing category.
- 5) Provide the total spending by customer type and identify which customer type spends more.
- 6) Provide the number of transactions by payment method and identify the most popular payment method.
- 7) Provide the average transaction value for the supermarket.
- 8) Provide the customer satisfaction percentage by branch and identify the branch with the highest satisfaction.
- 9) Provide the total revenue by day and by month and identify the day and month with the highest sales.
- 10) Provide the overall percentage of satisfied customers/transactions.